

Lab Setup – ElasticSearch / Kibana

1. Assumed Lab Environment

This manual assumes the following setup:

Item	Value
Host platform	VirtualBox
Guest OS	Ubuntu
Elastic version	9.4.2
Elasticsearch container name	es01
Kibana container name	kib01
Docker network	elastic
Elasticsearch URL	https://localhost:9200
Kibana URL	http://localhost:5601
Authentication	elastic user password
CA certificate	http_ca.crt copied from Elasticsearch container

2. Create a Working Folder

Open a terminal in Ubuntu:

```
mkdir -p ~/elastic-lab
cd ~/elastic-lab
```

This folder will hold:

- Elasticsearch CA certificate
- downloaded datasets
- import scripts
- verification commands

3. Install Docker Engine on Ubuntu

Install Docker Engine using Docker's official Ubuntu instructions.

<https://docs.docker.com/engine/install/ubuntu/>

After installation, verify Docker:

```
sudo docker version
sudo docker run hello-world
```

Expected result:

Hello from Docker!

Optional but recommended for convenience:

```
sudo groupadd docker
sudo usermod -aG docker $USER
newgrp docker
```

4. Create the Docker Network

```
cd ~/elastic-lab
sudo docker network create elastic
```

If the network already exists, Docker may show an error. That is not a problem. You can verify it using:

```
sudo docker network ls
```

5. Pull Elasticsearch

```
sudo docker pull docker.elastic.co/elasticsearch/elasticsearch:9.4.2
```

6. Start Elasticsearch

```
sudo docker run --name es01 \
  --net elastic \
  -p 9200:9200 \
  -it \
  -m 4GB \
  docker.elastic.co/elasticsearch/elasticsearch:9.4.2
```

Important:

- Do not close this terminal.
- On first startup, Elasticsearch prints:
 - the elastic user password
 - the Kibana enrollment token

Copy and save both. You will need them later.

7. Copy the Elasticsearch CA Certificate

Open a second terminal.

Go to the lab folder:

```
cd ~/elastic-lab
```

Copy the CA certificate:

```
sudo docker cp es01:/usr/share/elasticsearch/config/certs/http_ca.crt .
```

Fix file ownership.

```
sudo chown $USER:$USER http_ca.crt
chmod 644 http_ca.crt
```

Verify the file exists:

```
ls -l http_ca.crt
```

8. Set the Elasticsearch Password Variable

Use straight double quotes, not curly quotes:

```
export ELASTIC_PASSWORD="PASTE_THE_GENERATED_PASSWORD_HERE"
```

Verify that the variable is set:

```
echo $ELASTIC_PASSWORD
```

9. Verify Elasticsearch

Run:

```
curl --cacert http_ca.crt \
  -u elastic:$ELASTIC_PASSWORD \
  https://localhost:9200
Expected result:
{
  "name" : "es01",
  "cluster_name" : "docker-cluster",
  "cluster_uuid" : "...",
  "version" : {
    "number" : "9.4.2"
  },
  "tagline" : "You Know, for Search"
}
```

If this works, Elasticsearch is ready.

10. Pull Kibana

```
sudo docker pull docker.elastic.co/kibana/kibana:9.4.2
```

11. Start Kibana

Open a new terminal, run:

```
KIBANA_ENCRYPTION_KEY=$(openssl rand -hex 32)
sudo docker run --name kib01 \
--net elastic \
-p 5601:5601 \
-e XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY="$KIBANA_ENCRYPTION_KEY" \
docker.elastic.co/kibana/kibana:9.4.2
```

Open Kibana in the Ubuntu browser:

http://localhost:5601

To generate a new Kibana enrollment token from the running Elasticsearch container:

```
sudo docker exec -it es01 /usr/share/elasticsearch/bin/elasticsearch-create-enrollment-
token -s kibana
```

Paste the Kibana enrollment token when prompted.

Log in with username = elastic and password that Elasticsearch generated earlier.

If you forgot the password, reset it:

```
sudo docker exec -it es01 \
/usr/share/elasticsearch/bin/elasticsearch-reset-password \
-u elastic
```

12. Stopping and Restarting the Lab

Stop Kibana

```
sudo docker stop kib01
```

Stop Elasticsearch

```
sudo docker stop es01
```

Restart Elasticsearch

```
sudo docker start -ai es01
```

Keep this terminal open.

Restart Kibana

```
sudo docker start -ai kib01
```

Then open:

http://localhost:5601

If Containers Already Exist but Will Not Start

Check containers:

```
sudo docker ps -a
```

If needed, remove and recreate them:

```
sudo docker rm kib01
sudo docker rm es01
```

13. Add Data Samples

Download and save provided data samples to your home directory.

Before proceeding any further, turn off your machine and take a snapshot of the guest VM.
